

VCS Module

VMD0062

CO₂ CAPTURE FROM NATURAL GAS PROCESSING

Version 1.0

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Sectoral Scope 16: Carbon Capture and Storage

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CONTENTS

1	SUMMARY DESCRIPTION.....	5
2	SOURCES.....	5
3	DEFINITIONS.....	6
4	APPLICABILITY CONDITIONS	7
5	PROCEDURES	8
5.1	Module Boundary	8
5.2	Baseline Scenario	12
5.3	Quantification of Project Emissions	13
5.4	Quantification of Leakage.....	15
5.5	Uncertainty	16
6	DATA AND PARAMETERS.....	16
6.1	Data and Parameters Available at Validation	16
6.2	Data and Parameters Monitored.....	17
6.3	Description of Monitoring Plan.....	25
7	REFERENCES.....	26
	APPENDIX 1: JUSTIFICATION OF COMPATIBILITY WITH TRANSITION TO NET ZERO ...	28
	APPENDIX 2: DEMONSTRATING METHANE INTENSITY	29
	DOCUMENT HISTORY	31

1 SUMMARY DESCRIPTION

This module applies to activities that divert CO₂ separated from natural gas streams through acid gas removal processes, which would have been released to the atmosphere, to permanent geological storage. The module must be used in conjunction with VCS methodology *VM0049 Carbon Capture and Storage*.

This module establishes applicability conditions, defines the module boundary, and provides procedures and requirements for determining the baseline scenario, quantifying project emissions ($PE_{cap,y}$) and leakage emissions ($LE_{cap,y}$), and monitoring.

The project and leakage emissions calculated in this module, together with those calculated in the transport and storage modules, are used in *VM0049* to calculate the net GHG emission reductions (“reductions”) from projects using this module.

2 SOURCES

This module is used in combination with the most recent versions of *VM0049* and the following VCS Program modules and tools:

Capture Modules

- *VMD0056 CO₂ Capture from Air (Direct Air Capture)*
- *VMD0059 CO₂ Capture from Bioenergy*

Transport Module

- *VMD0057 CO₂ Transport for CCS Projects*

Storage Module

- *VMD0058 CO₂ Storage in Saline Aquifers and Depleted Hydrocarbon Reservoirs*

Other Modules, Tools, and Requirements

- *VT0010 Emissions from Electricity Consumption and Generation*
- *VT0012 Accounting non-VCS CO₂ in CCS Projects*
- *Geological Carbon Storage (GCS) Non-Permanence Risk Tool*
- *GCS Requirements*

This module uses the most recent versions of the following Clean Development Mechanism (CDM) tools:

- *TOOL09 Determining the Baseline Efficiency of Thermal or Electric Energy Generation Systems*
- *TOOL10 Tool to Determine the Remaining Lifetime of Equipment*

3 DEFINITIONS

In addition to the definitions set out in the *VCS Program Definitions* and *VM0049*, the following definitions apply to this module.

Acid gas

Hydrogen sulfide (H₂S) and/or carbon dioxide (CO₂) contaminants present in raw natural gas streams that are separated by an acid gas removal unit¹

Capture materials

The chemicals and media used to capture carbon dioxide. Depending on the technology, this may include capture solvents, solid sorbents, membranes, or catalysts, which may have to be replaced periodically due to loss or degradation over time.

Greenfield facility

A natural gas processing plant that began operations on or after 1 September 2025²

Methane emission intensity

The ratio, expressed in percent, of the total volume of hydrocarbon gas emissions from a facility in the monitoring period divided by the volume of hydrocarbon gas produced or processed at the facility during that period, for gas production and gas processing facilities respectively³

Natural gas processing plant

An industrial facility that processes raw natural gas to produce compressed, pipeline-quality natural gas and hydrocarbon by-products⁴

Natural gas production facility

An industrial facility that extracts raw natural gas streams from an underground deposit or reservoir.⁴

¹ Adapted from the US Environmental Protection Agency (US EPA): eCFR :: 40 CFR 98.238 – Definitions. Available at: <https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-98/subpart-W/section-98.238>

² This is the date on which Verra began development of this module.

³ Adapted from Environment and Climate Change Canada: Regulations Respecting Reduction in the Release of Methane and Certain Volatile Organic Compounds (Upstream Oil and Gas Sector), SOR/2018-66 (2018). <https://laws-lois.justice.gc.ca/eng/regulations/SOR-2018-66/page-1.html#h-858529>

⁴ Adapted from the US EPA's definition of natural gas processing: eCFR :: 40 CFR 98.230 – Definition of the source category. Available at: <https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-98/subpart-W/section-98.230>

4 APPLICABILITY CONDITIONS

This module applies to project activities that divert CO₂ separated from raw natural gas streams through acid gas removal processes, which would have been released to the atmosphere, to permanent geological storage.

This module is applicable under the following conditions:

- 1) Project activities include at least one of the following:
 - a) PA1: The project activity involves the installation and operation of new equipment or systems to divert CO₂ toward permanent storage where none was in place. This activity applies to existing and greenfield natural gas processing plants.
 - b) PA2: The project activity includes the refurbishment of existing equipment or systems that divert CO₂ toward permanent storage which are in place but have ceased operation. This activity applies to existing natural gas processing plants only.
 - c) PA3: The project activity expands the diversion capacity of an existing system which diverts CO₂ toward permanent storage. This activity applies to existing natural gas processing plants only.
- 2) Capture occurs using one or a combination of the following processes:
 - a) Chemical or physical absorption or adsorption, with liquid solvents or solid sorbents (e.g., amines)
 - b) Membrane processes
- 3) The capture process used in the project activities is designed to regenerate the primary capture fluid or media, such that it is not a one-time use, and a concentrated CO₂ stream is recovered from regeneration and available for subsequent transport (where applicable) and storage.

This module is not applicable under the following conditions:

- 4) Project activities comprise upgrades to natural gas processing facilities or changes in operational practices leading to improved capture efficiency.

5 PROCEDURES

5.1 Module Boundary

The module boundary includes at least the natural gas processing plant generating the CO₂ emissions captured and diverted for permanent storage by the project. Commonly used equipment and processes include:

- equipment used to generate airflow for the capture process;
- capture of CO₂ in contactors, beds, or vessels by absorption, adsorption, or other processes;
- regeneration processes to generate a CO₂ stream and recover capture fluid or media;
- conditioning of CO₂ to allow further processing of CO₂ along the carbon capture and storage (CCS) segments (namely transport and storage); and
- co-located utilities for the CO₂ capture process (e.g., air separation units, water treatment systems, electrical or steam systems).

Section 5 of the most recent version of VM0049 provides further details on determining the module boundary. The capture facility, ancillary sites, equipment, and relevant project emissions included in the module boundary and quantified using this module must be clearly identified and documented. The project proponent must ensure that equipment is not omitted or double-counted.

5.1.1 Non-VCS CO₂ in the Project Boundary

In cases where non-VCS CO₂ is captured in or flows through the project boundary, the most recent version of VCS tool *VT0012 Accounting non-VCS CO₂ in CCS Projects* must be used to calculate the proportion of project emissions and leakage emissions associated with that non-VCS CO₂.

For all projects using this module:

- CO₂ streams that captured and diverted from natural gas produced at a natural gas production facility utilizing extraction methods involving the injection of CO₂ (i.e., enhanced gas recovery) where the act of injection or the CO₂ injected was credited or accounted for in another project must be accounted for as non-VCS CO₂.

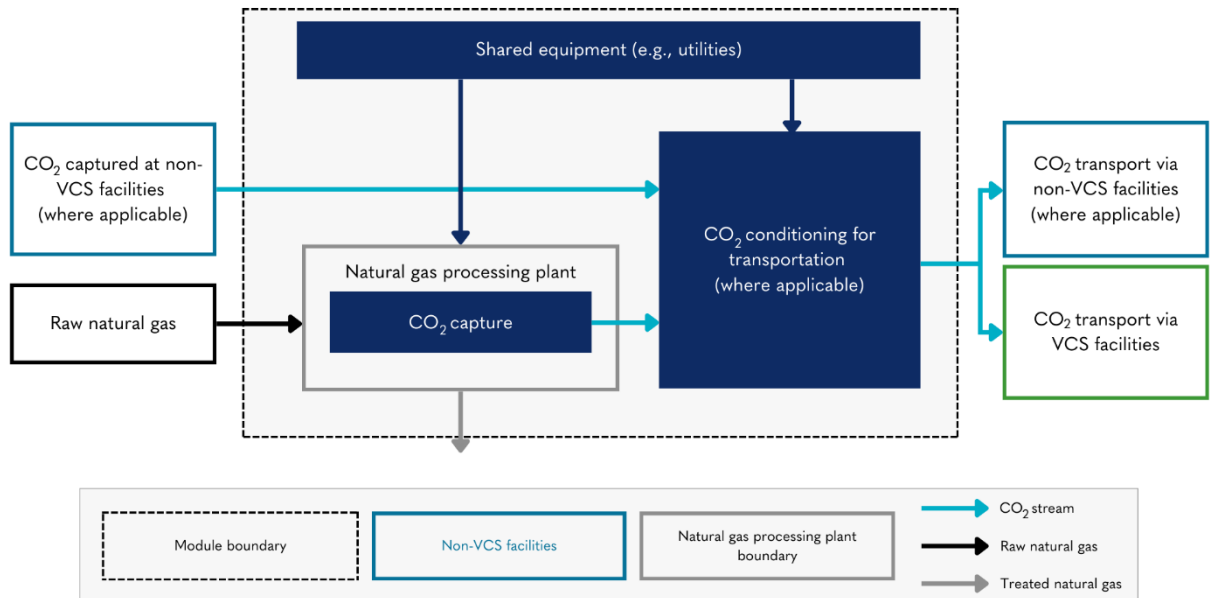
For projects involving greenfield natural gas processing facilities, in monitoring periods from 1 January 2030 onwards, the proportion of CO₂ in CO₂ streams that were captured and diverted from the following, must be accounted as non-VCS CO₂:

- natural gas produced at a natural gas production facility, where the facility cannot demonstrate a methane emission intensity of 0.2% or less

- natural gas processed at greenfield facilities, where the facility cannot demonstrate a methane emission intensity of 0.05% or less

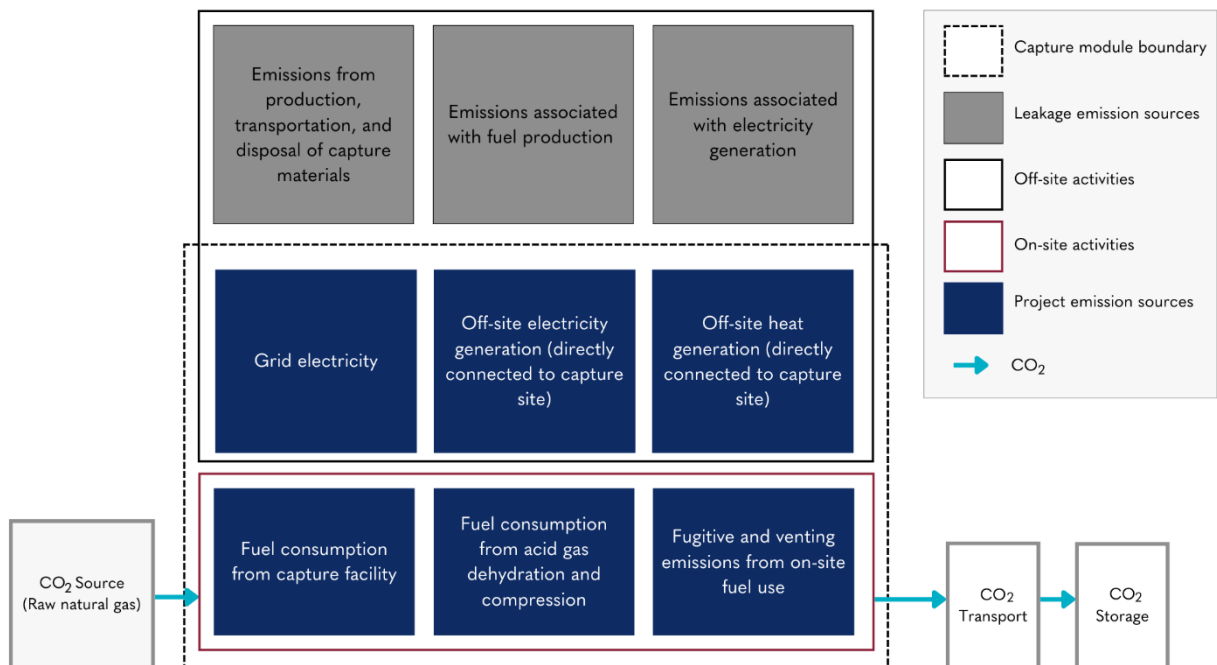
The boundary for this module is presented in Figure 1

Figure 1. Module boundary



Sources of GHG emissions within the module boundary are depicted in Figure 2.

Figure 2. Emission sources within the module boundary



The GHGs included in and excluded from the module boundary are given in Table 1.

Table 1. GHG sources and sinks accounted for as baseline, project, and leakage emissions

Source/Sink		Type	Gas	Included?	Justification/Explanation
Baseline	Fugitive and venting emissions from upstream gas production	Source	CO ₂	No	The components of the natural gas service present in the baseline are also present in the project and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Electricity consumption at capture facility	Source	CO ₂	No	The operation of the capture facility is consistent between the baseline and project scenarios and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Fuel consumption at capture facility	Source	CO ₂	No	The operation of the capture facility is consistent between the baseline and project scenarios and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Fugitive and venting emissions from on-site fuel use	Source	CO ₂	No	The components of the natural gas service present in the baseline are also present in the project and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Fuel combustion from flaring or incineration of acid gas or acid gas conditioning (e.g., sulfur recovery)	Source	CO ₂	No	Excluded for conservativeness ⁵
			CH ₄		
			N ₂ O		
			Other		
	Production, transportation, and disposal of capture materials	Source	CO ₂	No	The operation of the capture facility is consistent between the baseline and project scenarios and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		

⁵ Project proponents seeking credit for emissions avoidance from this source may apply a second methodology relevant to that activity, following Section 3.5 of the VCS Standard, v5.0.

Project	Electricity consumption from capture facility	Source	CO ₂	No	The operation of the capture facility is consistent between the baseline and project scenarios and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Fuel consumption from capture facility	Source	CO ₂	No	The operation of the capture facility is consistent between the baseline and project scenarios and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Fugitive and venting emissions from on-site fuel use for components not present in baseline	Source	CO ₂	No	De minimis
			CH ₄	Yes	Emission source
			N ₂ O	No	De minimis
			Other	No	No other greenhouse gases
	Electricity consumption for acid gas conditioning processes not present in baseline	Source	CO ₂	Yes	Emission source
			CH ₄	No	De minimis
			N ₂ O	No	De minimis
			Other	No	No other greenhouse gases
	Fuel consumption for acid gas conditioning processes not present in baseline	Source	CO ₂	Yes	Emission source
			CH ₄	Yes	Emission source
			N ₂ O	Yes	Emission source
			Other	No	No other greenhouse gases
Leakage	Fuel consumption for acid gas conditioning processes not present in baseline	Source	CO ₂	Yes	Emission source
			CH ₄	Yes	Emission source
			N ₂ O	No	De minimis
			Other	No	No other greenhouse gases
	Fugitive and venting emissions from upstream gas production	Source	CO ₂	No	The operation of the capture facility is consistent between the baseline and project scenarios and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		
	Production, transportation, and disposal of capture materials	Source	CO ₂	No	The components of the natural gas service present in the baseline are also present in the project and therefore the emissions are equivalent.
			CH ₄		
			N ₂ O		
			Other		

5.2 Baseline Scenario

Project proponents must accurately determine the activities and GHG emissions that would have occurred in the absence of the project activity.

5.2.1 Step 1: Identify Alternative Scenarios

Project proponents must assess which of the following alternative baseline scenarios is relevant to their project based on the classification of their project activity in Section 4.

Baseline scenario B1: CO₂ is vented, flared, or incinerated at a natural gas processing plant.

- a) Sub-scenario B1a: No equipment or systems that divert captured CO₂ toward permanent storage were in place at the natural gas processing plant before the operational start of project activities. This sub-scenario applies to project activity PA1.
- b) Sub-scenario B1b: Equipment or systems that divert captured CO₂ toward permanent storage were installed at the natural gas processing plant but had reached their end of life and ceased operation before the operational start of project activities. The most recent version of CDM *TOOL10 Tool to Determine the Remaining Lifetime of Equipment* must be applied to demonstrate that the remaining lifetime of the capture facility was zero before the operational start of project activities. This sub-scenario applies to project activity PA2.

Baseline scenario B2: CO₂ is captured and diverted toward permanent storage, and/or utilization in products or processes, from an existing natural gas processing plant at historical levels. In this scenario, equipment or systems that divert captured CO₂ were in place and operational before the operational start of project activities. This scenario applies to project activity PA3.

Baseline scenario B3: The proposed project activity occurs without being registered with a GHG program. This scenario could apply to PA1, PA2, and PA3.

5.2.2 Step 2: Check Consistency with Mandatory Applicable Laws and Regulations

Project proponents must use Step 1b of the most recent version of VCS tool VT0008 *Additionality Assessment* to identify the alternative scenarios from Step 2 (Section 5.2.1 in this module) that comply with all mandatory applicable legal and regulatory requirements within the applicable geographic region. All relevant policies and regulations in the applicable geographic region related to the project activity must be assessed, including those with objectives other than GHG emission reductions and/or carbon dioxide removals (e.g., for local air pollution control).

5.2.3 Step 3: Establish the Baseline Scenario

Where the only remaining alternative scenario is B3, the project activity represents the baseline scenario, and the project is not eligible to apply this module.

Otherwise, the remaining alternative B1a, B1b, or B2 is the baseline scenario and is used to inform the assessment of project additionality using Section 7 of VM0049.

5.3 Quantification of Project Emissions

The following sections provide guidance for determining the emissions that result from project activities (i.e., $PE_{Cap,y}$). Project emissions are calculated as per Equation (1).

$$PE_{Cap,y} = PE_{Comb_Fuel,y} + PE_{Fuel_FV,y} + PE_{Elec,y} - PE_{nonVCS\ CO_2,y} \quad (1)$$

Where:

- $PE_{Cap,y}$ = Project emissions from capture in year y (t CO₂e)
- $PE_{Comb_Fuel,y}$ = Project emissions from fuel combustion to operate acid gas dehydration, compression, and conditioning processes in year y (t CO₂e)
- $PE_{Fuel_FV,y}$ = Fugitive and venting from on-site natural gas use to operate equipment not present in the baseline in year y (t CO₂e)
- $PE_{Elec,y}$ = Project emissions from electricity consumption to operate equipment not present in the baseline in year y , calculated using VCS tool VT0010 *Emissions from Electricity Consumption and Generation* (t CO₂e)
- $PE_{nonVCS\ CO_2,y}$ = Project emissions from processes and equipment related to non-VCS sources in year y determined using the most recent version of VT0012; equal to zero for projects with no non-VCS CO₂ (t CO₂e)

5.3.1 Project Emissions from Fuel Combustion

Project emissions from fossil fuel combustion for acid gas dehydration, conditioning, and compression processes are calculated in Equation (2).

$$PE_{Comb_Fuel,y} = \sum_d (Q_{Fuel,d,y} \times EF_{Fuel,CO_2,d}) + (Q_{Fuel,d,y} \times EF_{Fuel,CH_4,d} \times GWP_{CH_4}) + (Q_{Fuel,d,y} \times EF_{Fuel,N_2O,d} \times GWP_{N_2O}) \quad (2)$$

Where:

- $PE_{Comb_Fuel,y}$ = Project emissions from fuel combustion to operate acid gas dehydration, compression, and conditioning processes in year y (t CO₂e)
- $Q_{Fuel,d,y}$ = Quantity of fuel type d used to operate on-site and/or third-party (for off-site heat/steam supply) equipment in year y (m³ or kg or GJ)
- $EF_{Fuel,CO_2,d}$ = Carbon dioxide emission factor for combustion of fuel d ; equal to zero for fuels derived from renewable biomass (t CO₂/m³ or t CO₂/kg or t CO₂/GJ)
- $EF_{Fuel,CH_4,d}$ = Methane emission factor for combustion of fuel d (t CH₄/m³ or t CH₄/kg or t CH₄/GJ)
- $EF_{Fuel,N_2O,d}$ = Nitrous oxide emission factor for combustion of fuel d (t N₂O/m³ or t N₂O/kg or t N₂O/GJ)
- GWP_{CH_4} = Global warming potential for methane (t CO₂e/t CH₄)
- GWP_{N_2O} = Global warming potential for nitrous oxide (t CO₂e/t N₂O)

Where an emission factor that considers only combustion emissions is not available for a given fuel, a combined emission factor for both combustion and upstream emissions may be used as $EF_{Fuel,d}$ in Equation (2).

Off-site Fuel Consumption

The quantity of power and/or heat supplied from a directly connected off-site facility is determined using Equation (3).

$$Q_{Fuel,d,y} = Q_{Cogen,d,y} \times \frac{\left(\frac{Heat_{AG_conditioning,y}}{\eta_{Heat,y}} + \frac{Electricity_{AG_conditioning,y}}{\eta_{Elec,y}} \right)}{\left(\frac{Heat_{Cogen,y}}{\eta_{Heat,y}} + \frac{Electricity_{Cogen,y}}{\eta_{Elec,y}} \right)} \quad (3)$$

Where:

$Q_{Fuel,d,y}$	=	Quantity of fuel type d used to operate on-site and/or third-party (for off-site heat/steam supply) equipment in year y (m ³ or kg or GJ)
$Q_{Cogen,d,y}$	=	Quantity of fuel type d used by the cogeneration unit to generate electricity and/or heat in year y (m ³ or kg or GJ)
$Heat_{AG_conditioning,y}$	=	Quantity of useful thermal energy supplied for acid gas conditioning by the cogeneration unit in year y ; equal to zero where only electricity is supplied to the capture facility (MWh)
$Electricity_{AG_conditioning,y}$	=	Quantity of electricity supplied for acid gas conditioning by the cogeneration unit in year y ; equal to zero where only heat is supplied to the capture facility (MWh)
$Heat_{Cogen,y}$	=	Total quantity of useful thermal energy produced by the cogeneration unit in year y (MWh)
$Electricity_{Cogen,y}$	=	Total quantity of electricity produced by the cogeneration unit in year y (MWh)
$\eta_{Heat,y}$	=	Efficiency of thermal energy production from cogeneration unit in year y determined using the most recent version of CDM TOOL09 <i>Determining the Baseline Efficiency of Thermal or Electric Energy Generation Systems</i>
$\eta_{Elec,y}$	=	Efficiency of electricity production from cogeneration unit in year y determined using the most recent version of CDM TOOL09

Waste Heat

Project emissions from the consumption of waste heat may be assumed to be zero for heat sources that meet the definition of waste heat in VM0049. Emissions associated with the generation of heat that does not meet the definition of waste heat must be accounted for in Equations (2) or (3) as appropriate.

5.3.2 Fugitive and Venting Emissions from On-Site Fuel Use

Projects that use natural gas on-site must quantify fugitive and venting emissions from components that are not present in the baseline, using Equation (4).

$$PE_{Fuel_{FV},y} = (\sum_n Count_{n,y} \times EF_n \times T_{n,y} \times 0.001 + \sum_m V_m) \times GWP_{CH_4} \quad (4)$$

Where:

$PE_{Fuel_{FV},y}$	= Fugitive and venting from on-site natural gas use associated with project activities in year y (t CO ₂ e)
$Count_{n,y}$	= Total number of components n in use at the facility during year y that are not present in the baseline (unitless)
EF_n	= Emission factor of fugitive emissions for component n (kg CH ₄ /hr/unit)
$T_{n,y}$	= Time for which component n is pressurized in year y (hr)
V_m	= Vented methane emissions for venting event m (t CH ₄ /event)
0.001	= Conversion from kilograms to tonnes

5.4 Quantification of Leakage

The leakage emissions from project activities are calculated as per Equation (5).

$$LE_{Cap,y} = LE_{Fuel,y} - LE_{nonVCS\ CO_2,y} \quad (5)$$

Where:

$LE_{Cap,y}$	= Leakage emissions from capture in year y (t CO ₂ e)
$LE_{Fuel,y}$	= Leakage emissions from fuel combustion to operate acid gas dehydration, compression, and conditioning processes not present in the baseline, in year y (t CO ₂ e)
$LE_{nonVCS\ CO_2,y}$	= Leakage emissions from processes and equipment related to non-VCS CO ₂ streams in year y determined using the most recent version of VT0012; equal to zero for projects with no non-VCS CO ₂ (t CO ₂ e)

5.4.1 Leakage Emissions from Fuel Consumption ($LE_{Fuel,y}$)

Upstream emissions related to fossil fuel consumption for acid gas dehydration, compression, and conditioning processes are calculated using Equation (6).

$$LE_{Fuel,y} = \sum_d (Q_{Fuel,d,y} \times EF_{Upstream_{Fuel,d}}) \quad (6)$$

Where:

$LE_{Fuel,y}$	= Leakage emissions from fuel combustion to operate acid gas dehydration, compression, and conditioning processes not present in the baseline, in year y (t CO ₂ e)
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- $Q_{Fuel,d,y}$ = Quantity of fuel type d used to operate on-site and/or third-party (for off-site heat/steam supply) equipment not present in the baseline, in year y (m^3 or kg or GJ)
- $EF_{Upstream_Fuel,d}$ = Emission factor for upstream sources related to fuel type d used in the module boundary ($t\ CO_2e/m^3$ or $t\ CO_2e/kg$ or $t\ CO_2e/GJ$)

Where power and heat are supplied from an off-site facility, $Q_{Fuel,d,y}$ must be determined as a proportion of the total fuel used to generate the total electricity and heat at the directly connected facility, using Equation (3).

Where a combined emission factor considering both combustion ($EF_{Fuel,d}$) and upstream emissions ($EF_{Upstream_Fuel,d}$) for a given fuel is used in Equation (2), Equation (6) is not required for that fuel.

5.5 Uncertainty

The primary sources of uncertainty identified for the project activities covered in this module are measurement error related to fuel combustion and the leakage emissions from the production of capture materials. All other potential sources of uncertainty are either considered de minimis or accounted for in the conservativeness of the default factors.

- Uncertainty of measurement error related to project emissions from fuel combustion for capture activities is considered de minimis where the metering equipment used for determining fuel volumes ($Q_{Fuel,d,y}$) uses a custody transfer meter or fiscal metering for the transaction.
- Uncertainty of measurement error from leakage emissions from the production of capture materials is considered de minimis where the magnitude of the leakage emissions is less than 2% of the net reduction or removal benefit of the project.

Where project proponents cannot demonstrate that the relevant condition above is met, that source of uncertainty must be included in their uncertainty assessment as described in Section 8.5 of VM0049.

6 DATA AND PARAMETERS

Additional data and parameters are defined in VM0049 and related tools (VCS and CDM) as applicable.

6.1 Data and Parameters Available at Validation

Data/Parameter	GWP_{CH_4}
Data unit	$t\ CO_2e/t\ CH_4$

Description	Global warming potential for methane
Equations	(2), (4)
Source of data	Version of IPCC Assessment Report required by <i>VCS Standard</i>
Value applied	See the most recent version of the <i>VCS Standard</i> .
Justification of choice of data or description of measurement methods and procedures applied	Required by the <i>VCS Standard</i>
Purpose of data	Calculation of project emissions
Comments	None

Data/Parameter	GWP_{N2O}
Data unit	t CO ₂ e/t N ₂ O
Description	Global warming potential for nitrous oxide
Equations	(2)
Source of data	Version of IPCC Assessment Report required by <i>VCS Standard</i>
Value applied	See the most recent version of the <i>VCS Standard</i> .
Justification of choice of data or description of measurement methods and procedures applied	Required by the <i>VCS Standard</i>
Purpose of data	Calculation of project emissions
Comments	None

6.2 Data and Parameters Monitored

Data/Parameter	$Q_{Fuel,d,y}$
Data unit	m ³ or kg or GJ
Description	Quantity of fuel type d used to operate on-site and/or third-party (for off-site heat/steam supply) equipment not present in the baseline, in year y
Equations	(2), (6)
Source of data	Fuel receipts/invoices or flow meter readings

	Where the project proponent does not have access to fuel receipts/invoices and flow meter readings, equipment specifications and conservative runtime information may be used to estimate fuel consumption.
Description of measurement methods and procedures to be applied	Measured from flow meters or calculated from fuel receipts/invoices. Otherwise, equipment specifications and conservative runtime information may be used to estimate fuel consumption.
Frequency of monitoring/recording	Continuous
QA/QC procedures to be applied	Measuring equipment (e.g., flow meters, weighing scale) must operate within the manufacturer's specified operating conditions and must be routinely calibrated, inspected, and maintained according to manufacturer specifications.
Purpose of data	Calculation of project and leakage emissions
Calculation method	Monthly fuel consumption is determined by summing the quantities from calibrated device readings or fuel receipts/invoices.
Comments	None

Data/Parameter	$EF_{Fuel,CO_2,d}$ $EF_{Fuel,CH_4,d}$ $EF_{Fuel,N_2O,d}$
Data unit	t CO ₂ /m ³ or t CO ₂ /kg or t CO ₂ /GJ t CH ₄ /m ³ or t CH ₄ /kg or t CH ₄ /GJ t N ₂ O/m ³ or t N ₂ O/kg or t N ₂ O/GJ
Description	Carbon dioxide emission factor for combustion of fuel <i>d</i> Methane emission factor for combustion of fuel <i>d</i> Nitrous oxide emission factor for combustion of fuel <i>d</i>
Equations	(2)
Source of data	The following data sources may be used: 1) Emission factor from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 2 Stationary Combustion, Table 2.2 ⁶ 2) Emission factors published by US EPA (2023) ⁷ or similar source 3) Data provided by the fuel supplier 4) Determination of combustion emissions using direct measurement methods, where the measurement approach is consistent with

⁶ Available at: https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf

⁷ Available at: <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>

	approved guidance from a recognized regulatory authority, such as the US EPA's Electronic Code of Federal Regulations, Title 40, Part 98, Subpart W, § 40 CFR Part 98, Subpart W §98.233(z) ⁸
Description of measurement methods and procedures to be applied	Use the most recent data published by the above sources when reporting project emissions.
Frequency of monitoring/recording	Annual
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	$EF_{Fuel,CO2,d}$ is equal to zero for fuels derived from sustainable biomass.

Data/Parameter	$Q_{Cogen,d,y}$
Data unit	m ³ or kg or GJ
Description	Quantity of fuel type d used by the cogeneration unit to generate electricity and/or heat in year y
Equations	(3)
Source of data	Fuel receipts/invoices or flow meter readings, as applicable
Description of measurement methods and procedures to be applied	Measured from flow meters or calculated from fuel receipts or invoices
Frequency of monitoring/recording	Continuous or for every invoice, aggregated annually
QA/QC procedures to be applied	Measuring equipment (e.g., flow meters, weighing scale) must operate within the manufacturer's specified operating conditions and must be routinely calibrated, inspected, and maintained according to manufacturer specifications.
Purpose of data	Calculation of project emissions
Calculation method	Monthly fuel consumption is determined by summing the quantities from calibrated device readings or fuel receipts/invoices.
Comments	Invoices and/or contracts with the third party must be in place to allow proper data collection.

⁸ Available at: [https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-98/subpart-W#p-98.233\(z\)](https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-98/subpart-W#p-98.233(z))

Data/Parameter	<i>Heat</i> _{AG_conditioning,y}
Data unit	MWh
Description	Quantity of useful thermal energy supplied for acid gas conditioning by the cogeneration unit in year y; equal to zero where only electricity is supplied to the capture facility
Equations	(3)
Source of data	Utility receipts/invoices or metered data for heat usage
Description of measurement methods and procedures to be applied	Measured from calorimeters or calculated from receipts/invoices, considering energy content in steam and condensate return as applicable based on steam properties
Frequency of monitoring/recording	Continuous, aggregated annually
QA/QC procedures to be applied	The calorimeter must be routinely calibrated, inspected, and maintained according to manufacturer specifications.
Purpose of data	Calculation of project emissions
Calculation method	Monthly supplied heat is determined by summing the quantities from calibrated device readings or fuel receipts/invoices.
Comments	Invoices and/or contracts with the third party must be in place to allow proper data collection.

Data/Parameter	<i>Electricity</i> _{AG_conditioning,y}
Data unit	MWh
Description	Quantity of electricity supplied for acid gas conditioning by the cogeneration unit in year y; equal to zero where only heat is supplied to the capture facility
Equations	(3)
Source of data	Utility receipts/invoices or metered data for electricity use
Description of measurement methods and procedures to be applied	Measured from electricity meters or calculated from receipts/invoices
Frequency of monitoring/recording	Continuous, aggregated annually

QA/QC procedures to be applied	Electricity meters must be routinely calibrated, inspected, and maintained according to manufacturer specifications.
Purpose of data	Calculation of project emissions
Calculation method	Monthly supplied electricity is determined by summing the quantities from calibrated device readings or fuel receipts/invoices.
Comments	Invoices and/or contracts with the third party must be in place to allow proper data collection. Value is equal to zero where only heat is supplied to the capture facility.

Data/Parameter	$Heat_{Cogen,y}$
Data unit	MWh
Description	Total quantity of useful thermal energy produced by the cogeneration unit in year y
Equations	(3)
Source of data	Utility receipts/invoices or metered data for heat produced
Description of measurement methods and procedures to be applied	Direct measurement of steam flows (or other heat transfer fluid) and characteristics at the cogeneration facility, considering energy content in steam and condensate return
Frequency of monitoring/recording	Continuous, aggregated monthly
QA/QC procedures to be applied	Calorimeters must be routinely calibrated, inspected, and maintained according to manufacturer specifications.
Purpose of data	Calculation of project emissions
Calculation method	Monthly heat production is determined by summing the quantities from calibrated device readings.
Comments	Invoices and/or contracts with the third party must be in place to allow proper data collection.

Data/Parameter	$Electricity_{Cogen,y}$
Data unit	MWh
Description	Total quantity of electricity produced by the cogeneration unit in year y
Equations	(3)
Source of data	Utility receipts/invoices or metered data for off-grid use

Description of measurement methods and procedures to be applied	Measured from electricity meters or calculated from receipts or invoices
Frequency of monitoring/recording	Continuous, aggregated monthly
QA/QC procedures to be applied	Electricity meters must be routinely calibrated, inspected, and maintained according to manufacturer specifications.
Purpose of data	Calculation of project emissions
Calculation method	Monthly electricity production is determined by summing the quantities from calibrated device readings.
Comments	Invoices and/or contracts with the third party must be in place to allow proper data collection.

Data/Parameter	$Count_{n,y}$
Data unit	unitless
Description	Total number of components n in use at the facility during year y that are not present in the baseline (unitless)
Equations	(4)
Source of data	Records of capture facility (e.g., pipe and instrument drawing, parts lists)
Description of measurement methods and procedures to be applied	Quantification is based on component counts and respective emission factors. Fugitive emissions are quantified following the approach in the US EPA's Electronic Code of Federal Regulations, Title 40, Part 98, Subpart W, § 98.233 (r).⁹ Examples of emission sources for fugitive emissions include components such as valves, pipe fittings/connectors, open-ended pipes, pressure relief valves, flanges, meters, and instruments.
Frequency of monitoring/recording	Annual
QA/QC procedures to be applied	Use the most recent data available from the capture facility.
Purpose of data	Calculation of project emissions
Calculation method	Annually count all n components in use at the facility from facility records (e.g., drawings, parts lists).

⁹ Available at: [https://www.ecfr.gov/current/title-40/part-98/subpart-W#p-98.233\(r\)](https://www.ecfr.gov/current/title-40/part-98/subpart-W#p-98.233(r))

Comments	Potential sources for fugitive emissions in the capture facility include components such as valves, pipe fittings/connectors, open-ended pipes, pressure relief valves, flanges, meters, and instruments.
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Data/Parameter	$T_{n,y}$
Data unit	hours
Description	Time for which component n is pressurized in year y
Equations	(4)
Source of data	Records of capture facility (e.g., control systems, recorded operational data)
Description of measurement methods and procedures to be applied	Option 1: Data from capture facility records Option 2: A default of 8760 hours, based on the assumption that components remain pressurized at all times throughout the year
Frequency of monitoring/recording	Annual
QA/QC procedures to be applied	Use the most recent data available from the capture facility.
Purpose of data	Calculation of project emissions
Calculation method	Use annual operational records to determine $T_{n,y}$ of each component n or use default value.
Comments	None

Data/Parameter	V_m
Data unit	t CH ₄ /event
Description	Vented CH ₄ emissions for venting event m
Equations	(4)
Source of data	Data from the capture facility
Description of measurement methods and procedures to be applied	Option 1: Direct measurement of venting Option 2: Estimated based on isolated volumes of pipes and equipment Option 3: Estimated based on non-isolated volumes of pipes and equipment. The project proponent must determine the quantity of vented CH ₄ by transient flow rate calculations for compressible fluids appropriate for the expected evolving conditions in the pipeline or component, based on the approximate geometry of the escaping flow and pipelines/components connected to the venting.

Frequency of monitoring/recording	Annual
QA/QC procedures to be applied	Cross-checked based on energy balance related to metered fuel use
Purpose of data	Calculation of project emissions
Calculation method	Determine V_m by direct measurement or estimate using isolated or non-isolated volumes of pipes and equipment. Use appropriate transient flow rate calculations for CH ₄ venting events.
Comments	None

Data/Parameter	EF_n
Data unit	kg CH ₄ /hr/unit
Description	Emission factor of fugitive emissions for component n
Equations	(4)
Source of data	Emission factor derived from subpart W of US EPA. 2023. <i>Mandatory GHG Reporting Program</i> ¹⁰ or equivalent nationally appropriate regulations
Description of measurement methods and procedures to be applied	Use the most recent data published by the above sources at the time of reporting project emissions.
Frequency of monitoring/recording	Annual
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	None

Data/Parameter	$EF_{Upstream_Fuel,d}$
Data unit	t CO ₂ e/m ³ or t CO ₂ e/kg or t CO ₂ e/GJ
Description	Emission factor for upstream sources related to fuel type d used in the module boundary

¹⁰ For more information see: <https://www.epa.gov/ghgreporting/subpart-w-petroleum-and-natural-gas-systems>

Equations	(6)
Source of data	1) An LCA conducted by a qualified third party in accordance with the most recent versions of ISO 14064 and 14067, that uses either primary or published and peer-reviewed data¹¹ 2) A compliance market-approved tool (e.g., CA-GREET,¹² GHGenius¹³) 3) Emission factors published in peer-reviewed literature¹⁴ representative of the fuels used in plant operation both temporally and geographically 4) Data provided by the fuel supplier or manufacturer where the data used are consistent with that reported to a regulatory body
Description of measurement methods and procedures to be applied	Use the most recent data published by the above sources when reporting project emissions.
Frequency of monitoring/recording	Annual
QA/QC procedures to be applied	Where peer-reviewed literature is used, it must have been published within a year of reporting project emissions and must be temporally and geographically representative of the capture facility.
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	None

6.3 Description of Monitoring Plan

6.3.1 Data Collection

The project proponent must design and implement a comprehensive and robust monitoring plan to ensure collection of all data and information required for each parameter listed in Sections 6.1–6.2.

¹¹ State or national government data on a fuel's carbon intensity are also acceptable sources of data for determining emission factors for fuels used by the project.

¹² The Greenhouse Gases, Regulated Emissions and Energy use in Technologies (GREET) model. Available at: <https://greet.es.anl.gov/>

¹³ Model for Life Cycle Assessment of Transportation Fuels. Available at: <https://www.ghgenius.ca/>

¹⁴ Peer-reviewed literature must be indexed in the Web of Science: Science Citation Index (SCI; available at <https://mil.clarivate.com>), as specified in the *VCS Methodology Requirements*.

6.3.2 Regulatory Limits on Carbon Dioxide Emissions

Project proponents must monitor and report any new or amended legal or regulatory framework that has been imposed and directly or indirectly limits GHG emissions from natural gas processing.

Where such new frameworks exist, the project proponent must reassess the additionality of the project using a project baseline based on the situation on the date on which the new framework was introduced, using Section 7 of VM0049.

6.3.3 Methane Emission Intensity

As of 1 January 2030, projects involving greenfield facilities must monitor the methane emission intensity of natural gas production and natural gas processing facilities relevant to the project, to inform the categorization of gas streams outlined in Section 5.1.1.

7 REFERENCES

American Carbon Registry. 2021. *Methodology for the Quantification, Monitoring, Reporting, and Verification of Greenhouse Gas Emissions Reductions and Removals from Carbon Capture and Storage Projects*. ACR, Winrock International. <https://acrcarbon.org/wp-content/uploads/2023/09/ACR-CCS-v2.0-Public-Comment-Version.pdf>

California Air Resources Board. 2018. *Carbon Capture and Sequestration Protocol under the Low Carbon Fuel Standard*. California Air Resources Board. https://ww2.arb.ca.gov/sites/default/files/2020-03/CCS_Protocol_Under_LCFS_8-13-18_ada.pdf

EU. 2009. “Directive 2009/31/EC of the European Parliament and of the Council of 23 April 2009 on the Geological Storage of Carbon Dioxide and Amending Council Directive 85/337/EEC, European Parliament and Council Directives 2000/60/EC, 2001/80/EC, 2004/35/EC, 2006/12/EC, 2008/1/EC, and Regulation (EC) No 1013/2006.” *Official Journal of the European Union*, 5.6.2009, L140–114-35.

EU. 2018. “Directive 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the Promotion of the Use of Energy from Renewable Sources.” *Official Journal of the European Union*, 21.12.2018, L328/82–L328/209.

European Commission. 2022. *Impact Assessment Report Part 1/2 and Part 2/2. Accompanying Documents to Proposal for a Regulation of the European Parliament and of the Council Establishing a Union Certification Framework for Carbon Removals*, EU Commission Staff Working Document 52022SC0377. European Commission. <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=SWD:2022:377:FIN>

Forbes, S. M., P. Verma, T. E. Curry, S. J. Friedmann, and S. M. Wade. 2008. *Guidelines for Carbon Capture, Transport and Storage*. World Resources Institute.
https://files.wri.org/d8/s3fs-public/pdf/ccs_guidelines.pdf

IETA. 2022. *High-level Criteria for Carbon Geostorage Activities. Provisional Version 1.0*. PowerPoint Presentation (ieta.org)

International Organization for Standardization. 2017. *ISO/TR27915:2017 Carbon Dioxide Capture, Transportation and Geological Storage – Quantification and Verification*. International Organization for Standardization.

IPCC. 2005. *IPCC Special Report on Carbon Dioxide Capture and Storage*. Prepared by Working Group III of the Intergovernmental Panel on Climate Change [edited by B. Metz, O. Davidson, H. C. de Coninck, M. Loos, and L. A. Meyer]. Cambridge University Press.
https://www.ipcc.ch/site/assets/uploads/2018/03/srccs_wholereport-1.pdf

US Department of Energy. 2022. *Best Practices for Life Cycle Assessment (LCA) of Direct Air Capture with Storage (DACs)*. <https://www.energy.gov/sites/default/files/2022-06/FECM%20DACs%20LCA%20Best%20Practices.pdf>

US EPA. 2023. GHG Emission Factors Hub. <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>

APPENDIX 1: JUSTIFICATION OF COMPATIBILITY WITH TRANSITION TO NET ZERO

The activities credited with this module serve to capture and sequester emissions at natural gas processing plants which would have otherwise been emitted. This module is unlikely to provide an incentive for prolonging or increasing consumption of natural gas, and does not provide an incentive that is significant enough to enable new fossil fuel production infrastructure. Further, for projects involving greenfield facilities after 2030, this module excludes gas streams produced or processed at facilities that do not meet methane emission performance thresholds. VMD0062 is therefore compatible with a transition to net zero.

Consumer Behavior

Consumers use natural gas out of necessity and there are few applications where alternative fuels can reliably displace this consumption. Consumers typically use natural gas because their existing appliances and space heating equipment are gas-fired. Most seek to minimize gas consumption to reduce higher costs.

Further, consumers are disconnected from the natural gas supply chain. They do not have the option to select a source for their gas or the information to decide based on the characteristics of the individual gas streams that are aggregated into the gas network. So, it is unlikely that consumers would start using natural gas, use more, or prolong their use into the future because natural gas processing incorporates carbon capture and storage, especially considering that this would increase costs.

Physical Infrastructure and Business Practice

Verra analyzed the economics of the natural gas supply chain to determine the extent to which VMD0062 will impact business practices and infrastructure. The economics of gas processing are dominated by tolling agreements and the sale of natural gas and natural gas liquids. Any carbon revenue generated by a project using VMD0062, though overcoming the barriers associated with capture, transport, and storage, would represent a small fraction of broader gas processing revenue by comparison. Accordingly, the revenue from selling the carbon credits generated by eligible activities would not be significant enough to incentivize the development of new extraction or production facilities.

APPENDIX 2: DEMONSTRATING METHANE INTENSITY

As of 1 January 2030, proponents of projects involving greenfield facilities are required to demonstrate the methane intensity of the natural gas production facilities that supply gas streams to the project and of the natural gas processing plants themselves.

To determine methane intensity, the volume of methane gas emissions from the production or processing facility are divided by the total volume of gas produced or processed at the facility on an annual basis.

For the purposes of this module, the volume of methane gas emissions from the production or processing facility must be determined using results from an inspection program or from an emissions monitoring system paired with an inspection program, that meets the requirements of one of the following:

- A regulatory program (e.g., Environment and Climate Change Canada's Enhanced Methane Regulation,¹⁵ the American Environmental Protection Agency's OOOOa,¹⁶ the EU Methane Regulation (EU 2024/1787)¹⁷)
- A voluntary reporting framework under an external body, that includes a defined inspection program and can quantify total facility methane emissions (e.g., the Oil and Gas Methane Partnership 2.0 (OGMP 2.0)¹⁸, MiQ¹⁹, Veritas²⁰)
- A voluntary program implemented by the oil and gas operator that publishes annual performance reports and meets the requirements in (1)–(2) below.
 - 1) For inspection programs:
 - a) Where an inspection program is being used without emissions monitoring systems, facilities are inspected at least quarterly.
 - b) The inspection program uses an instrument that:
 - i) is capable of measuring at least 500 ppmv of hydrocarbon,
 - ii) has a scale that is readable to ± 12.5 ppmv, and

¹⁵ <https://laws-lois.justice.gc.ca/eng/regulations/SOR-2018-66/page-1.html#h-858529>

¹⁶ <https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-60/subpart-OOOOa>

¹⁷ <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32024R1787>

¹⁸ <https://www.ogmpartnership.org/>

¹⁹ <https://miq.org/>

²⁰ <https://veritas.gti.energy/>

- iii) is operated and maintained in accordance with the manufacturer's recommendations where available, otherwise in accordance with industry standards and best practices
 - c) The inspection is performed by a qualified technician.
- 2) For emissions monitoring systems:
- a) The monitoring system is constantly operating, except for periods of maintenance.
 - b) Sensing equipment is:
 - i) capable of detecting methane flow rates of 1 kg/h or more,
 - ii) placed at locations where emissions are likely to occur within the facility, and
 - iii) calibrated in accordance with the manufacturer's recommendations where available, otherwise in accordance with industry standards and best practices.
 - c) Readings are taken at least once hourly and compiled annually.
 - d) Physical inspections take place at least annually and meet the criteria listed for inspection programs in (1)(b)–(c).

DOCUMENT HISTORY

Version	Date	Comment
v1.0	22 Dec 2025	Initial version